Sahej Singh

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Education

University of Illinois at Urbana-Champaign **Graduation: May 2026**
BS in Computer Engineering *GPA: 4.0*

Coursework: Artificial Intelligence, Principles of Safe Autonomy, Applied Parallel Programming, Analog Signal Processing, Communication Networks, Computer Systems Engineering, Algorithms and Models of Computation, Operating Systems, Digital Systems Laboratory, Data Structures, Discrete Structures, Linear Algebra with Computational Applications, Vector and Tensor Calculus, Statistics and Probability I & II

Experience

Haylon Technologies **Jun 2024 – Aug 2024**
Machine Learning Engineer Intern *Edison, NJ*

- Designed and deployed low-power **Machine Learning** models (SKLearn, TensorFlow) with online training, achieving 98% accuracy in predicting current spikes, reducing power wastage by 10%.
- Built a scalable **AWS** Lambda pipeline for real-time ML training on TimeStream data, cutting model update latency by 75% and enabling automated retraining with CI/CD.
- Optimized Particle Argon-based power monitoring system (C++, AWS IoT Core) using Logistic Regression, improving circuit power efficiency by 11%.

Autonomous and Unmanned Vehicle Systems Laboratory (AUVSL) **May 2023 – Dec 2024**
Research Intern *Urbana, IL*

- Integrated **CANBus** with **ROS2**, enabling real-time data exchange for autonomous vehicles, now used by 15+ researchers.
- Developed an adaptive control system using Lipschitz Neural Networks in ROS, enhancing vehicle stability in dynamic environments.
- Built a high-fidelity 6x6 vehicle simulation platform with **PyTorch** neural networks, enabling efficient model training and validation and improving loss from 6.5 to 0.785 (88% improvement).

Projects

CUDA CNN Model | *CUDA Git* **Jan 2024**

- Used NVIDIA's **CUDA** framework with GPU cluster access to implement **CNN** model from scratch.
- Optimized CNN inference using CUDA streams, shared memory, and DRAM coalescing, achieving a **1000x speedup** over CPU execution for large-scale ML workloads.

Neural Network Accelerator | *SystemVerilog, Vivado, FPGA Git* **May 2024**

- Designed and implemented an FPGA-based **deep learning accelerator**, optimizing fixed-point inference with custom memory controllers for efficient on-chip ML execution.
- Uses Block RAM IP core to store 8-bit **fixed-point data**, weights, biases for online training and classification of MNIST data on button press.
- Achieved **90%** MNIST classification accuracy with FPGA-accelerated online training, reducing compute latency.

AI-Powered Search & Recommendation System | *ML, Vector Databases, LLMs Git* **Jan 2025 – Present**

- Developed a **scalable ML-powered** ranking system, optimizing retrieval for **100K+** skincare product reviews with sub-500ms latency.
- Built a **real-time inference pipeline** using **Gemini LLM** embeddings & **Pinecone** vector search, improving query accuracy by **30%**.
- Designed & deployed a **cloud-based ML** inference system, cutting query processing time by **80%**, enabling seamless product recommendations.

Unix-Like Kernel | *C, RISC-V, GDB, QEMU* **Oct – Nov 2024**

- Implemented core OS functionalities (thread scheduling, virtual memory, system calls) in a Unix-like kernel (C, RISC-V), optimizing memory management for concurrent execution.
- Implemented thread **scheduling** (RR), device **drivers firmware** interface, a filesystem abstraction, program loading, **virtual memory**, processes, system calls, forking, and locking.
- Made extensive use of **GDB** for debugging and **Git** for version control.
- Developed multiple programs for users, including shell.

Skills

Programming: C, C++, Python, SystemVerilog, Rust
ML and Optimization: CUDA, TensorFlow, PyTorch, Fixed-Point Quantization, Model Compression
Hardware and Embedded : CUDA, TensorFlow, PyTorch, Fixed-Point Quantization, Model Compression
OS and Distributed Systems: Linux, QEMU, RISC-V, Docker, AWS Lambda